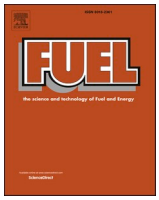
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Review article



Sustainable aviation fuel from waste cooking Oil: Pathways, Catalysts, and Prospects

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ABSTRACT

Faced with the urgent need for deep decarbonization in the aviation sector, Sustainable Aviation Fuel (SAF) derived from waste cooking oil (WCO) has emerged as a highly promising pathway. This review systematically examines the two most prominent technological routes for converting WCO into SAF: the commercially established Hydroprocessing of Esters and Fatty Acids (HEFA) process and the emerging Catalytic Hydrothermolysis Jet (CHJ) process. Central to both pathways is the role of catalysts, which govern reaction efficiency, product distribution, and overall process economics. This paper provides a comprehensive analysis of recent advances in catalyst design, with a particular focus on evaluating the impact of key process parameters (temperature, pressure, hydrogen-to-oil ratio, and liquid hourly space velocity) on reaction pathways, product selectivity, and catalyst stability. Despite persistent challenges in catalyst durability, feedstock impurities, and process integration, WCO-based SAF demonstrates significant lifecycle carbon reduction potential, abundant feedstock availability, and avoidance of food fuel competition. With ongoing technological improvements and strong policy support, WCO-to-SAF conversion holds substantial promise for advancing the green transformation of the aviation industry.

1. Introduction

1.1. Background

Given the excessive greenhouse gas emissions and the aviation industry's heavy reliance on traditional petroleum jet fuel, sustainable development has become a global concern. Decarbonization and emission reduction stand out as major challenges for this sector. Global initiatives are underway to realize carbon neutrality across all industries by 2050. Although aviation accounted for only 2–3% of global carbon emissions in 2019, its CO_2 emissions are projected to rise with growing aviation demand[1]. Furthermore, aviation activities generate non-CO_2 emissions, such as climate-damaging ozone and contrails[2]. Alternative aviation fuels, particularly Sustainable Aviation Fuel (SAF), which can significantly reduce both greenhouse gas (GHG) and non-CO_2 emissions, are being intensively developed to replace conventional aviation fuels and are a key pathway to achieving carbon neutrality goals[3].

The increase in human consumption of plant and animal oils has led to the rise in waste cooking oil (WCO) production. Rather than being disposed of, WCO can be technologically converted into a sustainable feedstock for biofuel production, thereby maximizing the value of biowastes through energy recovery while addressing disposal issues[4,5]. China ranks among the world's most abundant sources of WCO, with an estimated annual production of approximately 10 million tons[6]. However, due to an underdeveloped recycling system, the standardized recovery rate remains only about 50%. This waste cooking oil primarily originates from the catering industry and household discharges. Its main components are triglycerides (TG) and fatty acids (FA)[7], resulting in elevated acid value and oxygen content. Table 1 provides a comparative analysis of the key characteristics of waste cooking oil and fresh cooking oil. During the frying process, WCO becomes contaminated with various derivatives, including free fatty acids (FFA), volatile compounds (e.g.,

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low molecular weight alkanes, ketones, aldehydes, and acids), monomers, dimers, heterocycles, Maillard reaction products, and trace metals leached from packaging and food[8,9]. Consequently, WCO is unsuitable for human consumption, and its direct reuse poses significant health risks[10]. Due to its wide availability, significant carbon reduction potential, and non-competition with food supplies, WCO is considered an ideal feedstock for Bio-jet fuel. As the world's largest producer of WCO, the efficient resource utilization of WCO in China is of paramount importance for both energy security and environmental governance.

Given the high acid value, oxygen content, and complex impurities in waste WCO, its efficient conversion into SAF that meets stringent aviation standards is highly dependent on advanced catalytic processes. Currently, the most widely adopted technological route, Hydrogenation of Esters and Fatty Acids (HEFA), is centered on catalytic steps including hydrodeoxygenation (HDO), hydrocracking, and isomerization. Furthermore, emerging technologies like Catalytic Hydrothermolysis Jet (CHJ) also require hydrogenation upgrading, which similarly employs catalysts for hydroisomerization reactions. Catalyst performance directly impacts fuel yield, product quality, and process economics. It represents a core technological bottleneck in scaling up the conversion of WCO to SAF. This field currently has high technical barriers and a concentrated global supplier base, with companies such as BASF, Topsoe, and Albemarle actively developing specialized catalysts to meet the growing demand for SAF production. Therefore, in-depth research and optimization of catalyst systems are crucial for improving WCO conversion efficiency, reducing costs, and achieving deep decarbonization in the aviation sector.

According to a 2024 market report, global Bio-jet fuel production capacity reached approximately 2.3 million tons, with a significant concentration in the Asia-Pacific region[11]. Europe also leads in the research, development, and industrialization of SAF technologies. Europe also leads in the R&D and industrialization of SAF technologies. Finland's Neste leverages its proprietary NEXBTL hydrogenation technology, relying on robust sulfide catalysts (such as NiMo and CoMo) for deoxygenation. This enables flexible processing of diverse feedstocks like waste oils and fats. Its large-scale production is built upon a refined raw material supply chain supported by over 90,000 collection points worldwide. France's TotalEnergies provides a model for process integration and hydrogen management. Its Grandpuits biorefinery not only produces SAF but also incorporates a biomass-based hydrogen production unit. This facility utilizes refinery byproducts to generate renewable hydrogen, creating a closed-loop energy system that directly addresses the substantial hydrogen demand of the hydrodeoxygenation process. Meanwhile, its Normandy plant employs co-processing technology using waste cooking oil to produce SAF, and its Antwerp refinery employs co-processing to produce SAF, demonstrating a process configuration strategy for decarbonizing existing refining assets. China has made rapid strides in this field, driven primarily by breakthroughs in catalyst and process technologies. Sinopec has commissioned a 100,000-ton-per-year bio-aviation fuel plant that primarily uses WCO as feedstock, marking China's achievement in bridging the entire chain from technological R&D to commercial application. This process employs Sinopec's proprietary SRJET technology, whose key innovation lies in establishing a "two-stage, three-zone" molecular restructuring reaction system. It also involves the creation of highly stable hydrogenation-deoxygenation catalysts and highly selective alkane hydroisomerization catalysts. The Bio-jet fuel produced by this technology has obtained the international Responsible Sustainable Biofuels (RSB) certification and successfully completed a test flight in 2024 using a 30% Bio-jet fuel blend on the domestically produced C919 passenger aircraft. Furthermore, cases such as EcoCeres' HEFA plant supplying Bio-jet fuel to Europe, Shandong Sanju Biotechnology becoming China's third enterprise to obtain airworthiness certification, and the successful first-time startup of a 500,000-ton-per-year industrial facility utilizing specialized hydrogenation-isomerization catalysts and integrated technology developed by the Dalian Institute of Chemical Physics, Chinese Academy of Sciences, collectively demonstrate China's pivotal industrial breakthroughs in SAF catalyst systems and end-to-end process integration.

Policy support is a key driver for SAF adoption. Under the ReFuelEU Aviation initiative, starting January 1, 2025, all aviation fuel supplied at EU airports must contain a minimum of 2% SAF, rising to 6% by 2030 and 70% by 2050[12]. Similarly, the UK House of Commons has approved the Renewable Transport Fuel Obligation (Sustainable Aviation Fuel) Order 2024, which is expected to provide the UK aviation industry with access to approximately 1.2 million tons of SAF annually. In China, policy frameworks like the New Mechanism for Green and Low-Carbon Energy Transition explicitly support the R&D and industrial application of SAF technology, providing institutional safeguards for the aviation sector's low-carbon transition. According to the International Air Transport Association (IATA), over 45 airlines worldwide have now used Bio-jet fuel. Driven by these synergistic policy and market forces, a global SAF supply chain is rapidly emerging.

This paper reviews the research progress and industrial development of SAF derived from WCO, focusing on mainstream preparation technologies and key catalytic reactions, as well as the design, performance optimization, and industrial application of core catalysts. By systematically synthesizing relevant literature, policies, and industrial practice data, this work aims to clarify the technical bottlenecks and challenges in current WCO-to-SAF conversion pathway and to outline critical directions for future technological innovation and industrial upgrading.

1.2. *Bibliometric Analysis:Research Frontiers and emerging directions*

The data for this analysis were sourced from the Web of Science Core Collection database. Literature published between 2015 and 2025 was retrieved through a thematic search using the query: TS= (("waste cooking oil") AND ("Sustainable Aviation Fuel" OR "Bio-jet fuel")). The

Table 1
Comparative analysis of main physical and chemical properties of waste cooking oil and normal cooking oil[7,8].

Properties	Normal cooking oil	Waste cooking oil	Determination method	Reference standard
Acid value(mg KOH/g)	≤0.06	1.7–65.5	Potassium hydroxide titration	GB 2716–1988
Moisture content (wt. %)	≤0.05 ~ 0.2	5 ~ 30	Oven method/Karl Fischer method	GB 1537–2003
Density at 20 °C (kg/m3)	898–920	900–930	Densimeter method/pyknometer method	GB/T 5526–1985
Viscosity at 40 °C (mm2/s)	20–50	30–45	Capillary viscometer method	ASTM D445-14
Lodine value (cg/g)	≤0.05%(Mechanical impurities)	54–117	Gravimetric method/visual method	/
Peroxide value (meq/kg)	≤10	≥10	Sodium thiosulfate titration	GB/T 5538–2019-478
Saponification value (mgKOH/g)	190 ~ 200	Out of the normal range	Hydrochloric acid back titration after saponification reaction	/
Heavy metal(mg/kg)	Pb ≤ 0.1 ; As ≤ 0.1	Lead, arsenic, copper, etc. significantly exceeded the standard	AAS/ICP-MS	GB 2762–2017
Oxygen content(%)	10%-12%	≥10 ~ 20	Element analysis/infrared spectroscopy/OSI	/

resulting literature records were downloaded in plain text format as "full records and cited references". After deduplication and filtering using CiteSpace software (version 6.4.2), 507 valid documents were obtained, which helped to outline the research trajectory and cutting-edge developments in the conversion of waste cooking oil into SAF and related biofuels.

The keyword co-occurrence network (Fig. 1) illustrates the research landscape and core topics within this field. The network diagram comprises 293 nodes and 1,222 connections, with a network density of 0.0286. The co-occurrence network reveals that "waste cooking oil", "biodiesel" and "catalyst" form a tightly interconnected core, while the prominence of keywords like "Sustainable Aviation Fuel" and "hydrodeoxygenation" reflects early research focusing on the relatively mature technology pathway of converting WCO into biodiesel. However, the significant emergence of keywords like "Sustainable Aviation Fuel" and "hydrogenation deoxygenation" signals a clear shift in research focus toward producing high-value aviation fuels and their core advanced hydrogenation technologies.This technological divergence is quantitatively confirmed by cluster analysis (Fig. 2). The high modularity index ($Q = 0.6059$) indicates the emergence of multiple closely connected yet thematically distinct academic communities within the field. Among these, the "biodiesel" and "diesel engine" clusters represent traditional fuel utilization pathways, while the distinct "hydrodeoxygenation" and "hydrocracking" cutting-edge technology clusters precisely correspond to the core reaction steps of the HEFA and CHJ processes for producing aviation fuel, which will be discussed in detail later in this paper.

In summary, bibliometric analysis clearly illustrates the trend in this field toward evolving from a broad "raw materials-processes" binary framework to a multidimensional sustainability evaluation system integrating "raw materials-processes-performance-economics-environment". The research focus is shifting from meeting basic fuel demands to overcoming technical bottlenecks in challenging decarbonization sectors such as aviation. The following discussion will follow this evolutionary trajectory, first analyzing the principles and processes of the two mainstream aviation fuel production pathways applicable to WCO (HEFA and CHJ). It will then delve into their core, the design, optimization, and challenges of the catalyst system, which is precisely the central issue currently being addressed by the research cluster.

2. Principles and processes of Bio-jet from waste cooking oil

2.1. Overview of waste cooking oil

2.1.1. Physical and chemical properties of waste cooking oil

Waste cooking oil is a complex mixture composed of fats and oils whose physicochemical properties differ significantly from those of refined vegetable oils, primarily due to high acid value, high viscosity, and a complex composition of impurities. WCO exhibits elevated acid values due to the hydrolytic production of free fatty acids (FFA), which degrade flow properties, cause equipment corrosion, and accelerate oxidative side reactions[13]. Moreover, waste cooking oil contains high oxygen content due to the glycerol skeleton and acyl part of the molecule[14]. This high oxygen content in the feedstock for aviation fuels results in poor compatibility with fossil fuels and increased viscosity (limiting blending potential and worsening cold flow properties), as well as high reactivity and poor stability. These factors contribute to polymerization, leading to the formation of gums and sludge, as well as reduced thermal and storage stability[15,16]. Despite these challenges, WCO has a high calorific value, and its low sulfur content and renewable nature make it a significant potential feedstock for producing SAF. However, it requires a series of processing steps to achieve the standard performance of conventional aviation fuel[17,18].

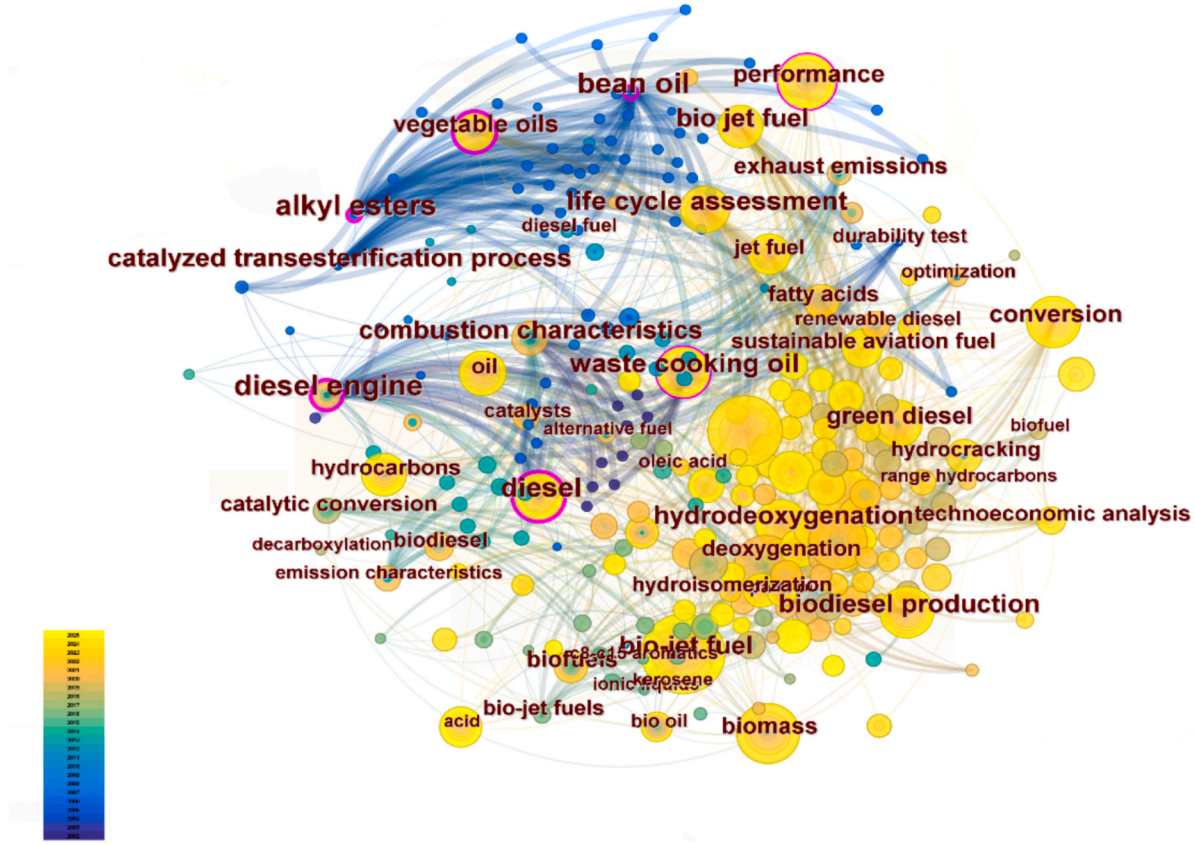


Fig. 1. Keyword co-occurrence analysis.

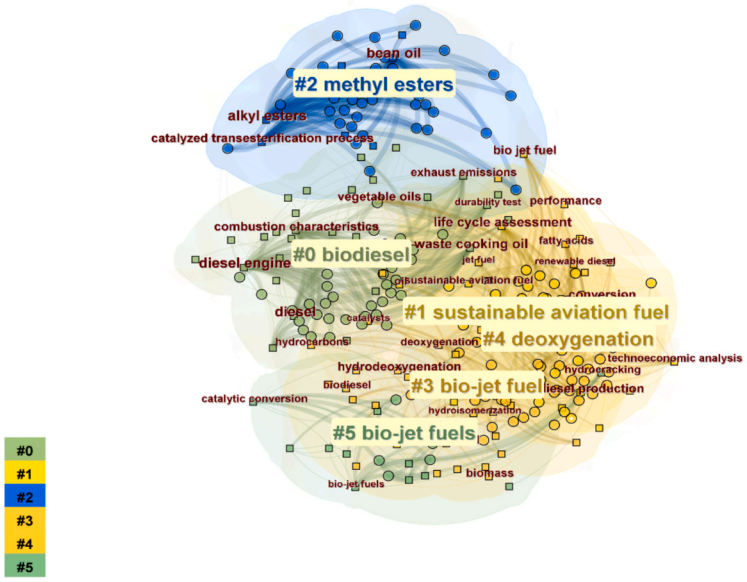


Fig. 2. Cluster analysis.

2.1.2. Waste cooking oil used as Sustainable aviation fuel raw material

The feedstock system of Bio-jet fuel is diversified and can usually be divided into three main generations[19]. The first generation of feedstocks are mainly food crops (e.g. corn, sugar cane); the second generation of feedstocks covers non-food energy crops (e.g. jatropha, castor) and waste biomass (e.g. waste cooking oil, agricultural and forestry residues); and the third generation of feedstocks focuses on algae resources[19].

Waste cooking oil represents a leading candidate among second-generation feedstocks, demonstrating considerable potential for sustainable energy production. Its current large-scale applications have been primarily in biodiesel production. However, in the field of SAF, WCO is becoming one of the core feedstocks for the hydrogenation (HEFA) route, due to its well-established supply chain and significant carbon reduction potential[21]. Hydrotreating technologies are being revolutionized globally, with the development of new reaction systems such as the "one-pot method", which aims to reduce energy consumption by about 15–20% compared to the traditional two-stage hydrotreating process (hydrodeoxygenation-hydroisomerization/cracking) by integrating multiple reaction steps[22]. Meanwhile, key qualities such as yield and cold flow performance of Bio-jet fuel are also being improved through high-throughput screening and rational design optimization of catalysts (e.g., development of non-precious-metal catalysts, modulation of carrier acidity, etc.)[21]. From a full life cycle perspective, waste cooking oil shows multi-dimensional advantages in Sustainable Aviation Fuel production:

(1) Resource recycling and sustainability: WCO is generated from kitchen waste, which has a huge annual potential, and the standardized recycling rate of used cooking oil in China is even less than 10% [23]. Converting it into aviation fuel through resource utilization can effectively solve the "waste oil disposal dilemma" and food safety problems, and avoid competing with food crops for arable land resources, which is a perfect fit for the strategic requirements of the circular economy and low-carbon transformation[20].

(2) Economic viability: Raw material costs are key to the economics of Bio-jet fuel. WCO's raw material costs can be significantly lower than conventional vegetable oils such as palm oil and soybean oil, and because it is derived from waste, it requires no additional land cultivation or agricultural inputs, resulting in a significant life-cycle cost advantage[24].

(3) In terms of environmental benefits, WCO-derived aviation fuel can reduce CO_2 by over 80% throughout its entire lifecycle compared to coal-to-jet fuel (CJF) [25–27]. In addition, WCO fuels have ultra-low sulfur characteristics (less than 1 ppm sulfur), much lower than conventional aviation kerosene (about 500–1000 ppm)[28]. During combustion, WCO-based fuels have a higher hydrogen to carbon ratio (H/C) and the concentration of aromatic hydrocarbon compounds is substantially reduced through hydrotreating processes, which effectively eliminate oxygenated functional groups and saturate unsaturated carbon bonds, resulting in a final hydrocarbon product with lower particulate matter (PM) emissions and a significant improvement in air quality in the airport area[21].

2.2. Preparation process for conversion of waste cooking oil to Bio-jet fuel

According to the latest ASTM D7566 standard (as of early 2026), the internationally recognized production pathways for Sustainable Aviation Fuel are currently categorized into eight major types, including Hydroprocessed Esters and Fatty Acids (HEFA), Fischer-Tropsch (FT), Alcohol-to-Jet (AtJ), Synthesized Iso-Paraffins (SIP), etc [26,29]. These pathways each possess distinct characteristics in terms of feedstock adaptability, technological maturity, carbon emission intensity, and economic viability[30]. However, for liquid biomass feedstocks like waste cooking oil, which exhibit high oxygen content, high acid value, and complex composition, the HEFA and CHJ pathways demonstrate unique advantages and industrial scalability[31]. Among these, HEFA is currently the only SAF production technology using oils and fats as feedstock that has achieved large-scale industrial application. Meanwhile, CHJ, as an emerging technology, is undergoing extensive experimental research due to its low hydrogen consumption and high-quality Bio-jet fuel characteristics. Although both HEFA and CHJ convert fatty acids and fatty acid esters from feedstocks into jet fuel, their conversion pathways differ: The HEFA process converts esters and/or fatty acids into hydrocarbons through dehydrogenation, hydrocracking, and hydroisomerization[16,32]. The CHJ process relies on hydrothermal liquefaction reactors and hydrogenation reactions to produce kerosene [32,33]. The HEFA pathway is recognized as the most technologically mature and commercially established approach sanctioned by the

American Society for Testing and Materials (ASTM) for generating Sustainable Aviation Fuel[16]. The CHJ is the sixth pathway approved by ASTM to produce Sustainable Aviation Fuel (SAF) in 2020.

2.2.1. Hydroprocessing of esters and fatty acids (HEFA)

The typical process flow of the HEFA process is shown in Fig. 3. Firstly, raw materials such as waste cooking oil are pretreated by physical filtration, dehydration and deacidification to remove impurities and free fatty acids. Subsequently, the pre-treated oils and fats are subjected to catalytic hydrogenation over catalysts (e.g., H-ZSM-5)[34] at high temperatures (250–400°C) and high pressures (3–10 MPa). This process primarily converts saturated fatty acid triglycerides into C_{15}-C_{18} straight-chain alkanes (i.e., hydrogenated vegetable oil) via the hydrodeoxygenation (HDO) pathway, simultaneously producing water and propane byproducts. To meet aviation fuel performance requirements, the resulting straight-chain alkanes require further hydroisomerization and selective cracking[35]. The final product is a bio-aviation fuel (C_8-C_{16}) compliant with ASTM D7566 standards[36], where the precise carbon number distribution can be achieved by modulating the catalyst acidity, metal active sites, and process parameters (e.g., temperature and pressure).

In terms of mechanism, the process is essentially a coupling of the targeted removal of oxygen atoms with carbon skeleton remodeling and is roughly divided into two steps. The first step is the conversion of double bonds in fatty acid chains to saturated fatty acids by catalytic hydrogenation with the addition of molecular hydrogen (Fig. 4a)[16], followed by the hydrodeoxygenation, decarboxylation, and decarbonylation of triacylglycerols. Hydrogenation of triacylglycerols is usually realized by β-elimination (Fig. 4b), and in addition to this it also may be realized by two different reaction pathways (γ-hydrogen migration, β-elimination and direct deoxygenation) (Fig. 4b)[37]. Within the decarboxylation pathway, the carboxylic acid moiety is cleaved from the triglyceride chain by direct scission of the C–C bond under H_2 atmosphere, and feedstock deoxygenation is achieved by the elimination of O_2 and the release of the byproduct, CO_2 (Fig. 4c). Analogously, in the decarbonylation route, the carboxyl moiety is cleaved from the triglyceride chain through direct C–C bond scission under H_2 atmosphere, leading to feedstock deoxygenation via O_2 elimination and the release of CO and H_2O as characteristic by-products (Fig. 4c)[16].

The second step involves hydroisomerization and hydrocracking reactions, where the feedstock from the first step is converted into straight-chain alkanes with carbon chain lengths exceeding C_{16}. N-alkanes usually have the disadvantages of low octane number (poor gasoline detonation resistance) and high freezing point (poor low-temperature fluidity in diesel and lubricating oils), and in order to improve the quality of the fuels, it is necessary to subject the generated hydrocarbons to hydrocracking and hydroisomerization treatments [38,39]. The hydrocracking and hydroisomerization reactions are typically carried out over a bifunctional catalyst and Fig. 5 shows the reaction mechanism of hydroisomerisation and hydrocracking of n-alkanes. The n-alkane molecules adsorb onto the metal sites of the catalyst to undergo dehydrogenation, losing two hydrogen atoms (H_2) to form the corresponding n-alkenes (A).

The generated alkene molecule diffuses away from the metal site and adsorbs onto a neighboring acidic site. A proton (H^+) adds to the alkene double bond, forming a high-energy n-alkyl carbonium ion (B). Side reactions also occur during this step: n-alkanes directly accept protons at acidic sites to form n-alkyl carbonylate ion (C); subsequent dehydrogenation yields n-alkyl carbonium ion (D). At this point, the reaction enters the core transformation pathway. The straight-chain n-alkyl carbenium ion undergoes a skeletal rearrangement via hydrogen migration or methyl migration, yielding a branched alkyl carbenium ion (monobranched, dibranched, tribranched) (F). When the branch position of alkyl carbenium ion (especially a multi-branched one) satisfies the conditions, C–C bond cleavage (β-cleavage) occurs, yielding x-alkenes and (n-x)-branched alkyl carbonium ion (G). The x-alkenes formed by cracking undergo hydrogenation at the metal site to produce a short-chain alkane (meeting aviation fuel carbon number requirements) (E). The (n-x)-branched alkyl carbonium ion undergoes "deprotonation" to form a branched-chain alkene, which is subsequently hydrogenated at the metal site to yield a branched-chain alkane (H)[38,40].

At present, although the technology of preparing Bio-jet fuel from bio-oils and fats through the traditional two-step process (hydrodeoxygenation to generate straight-chain alkanes, followed by isomerization/cracking) has been industrialized globally, its inherent defects, such as a long process flow, high reactor investment cost, and high consumption of hydrogen, have significantly raised the production cost. For this reason, the one-pot method, with its potential for simplified processes and energy savings, has become the focus of research. The core of this technology lies in the use of a multifunctional catalyst that couples multiple reaction steps, such as hydrodeoxygenation, hydroisomerization and selective cracking, in a single reactor. As shown in Fig. 6, the refined animal and vegetable fats and oils undergo hydrogen saturation, hydrodeoxygenation, alkane isomerization, and selective cracking under the action of bifunctional catalysts, and the standardized Bio-jet fuel are obtained directly through fractional distillation. The success of this process is critically dependent on the design of a high-performance bifunctional catalyst that enables simultaneous hydrodeoxygenation, cracking and isomerization activities, as well as good stability and anti-coking properties. This process is mainly in the research and development stage due to the challenging catalyst design, susceptibility to inactivation and the complexity and difficulty of the reaction network.

2.2.2. Catalytic Hydrothermolysis jet (CHJ)

The CHJ process for jet fuel represents a complete technological

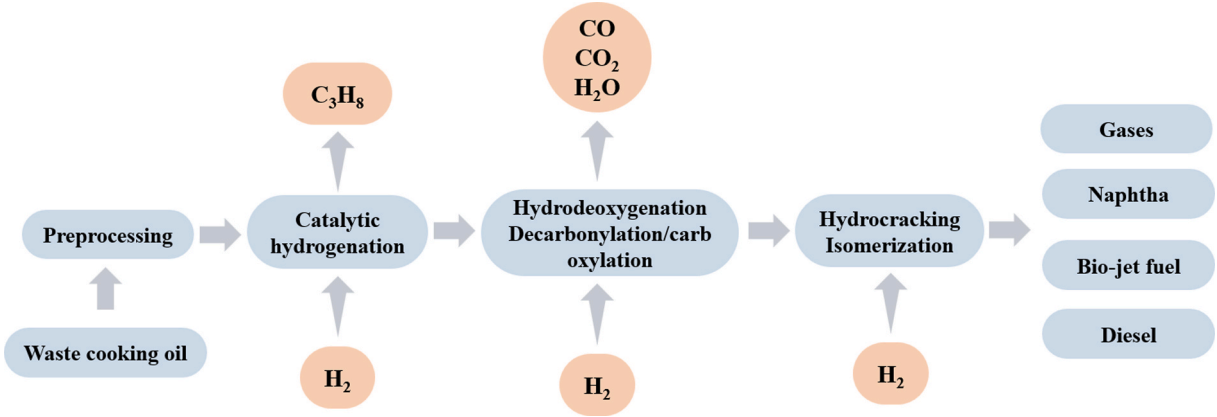


Fig. 3. Flow chart of ester and fatty acid hydrogenation.

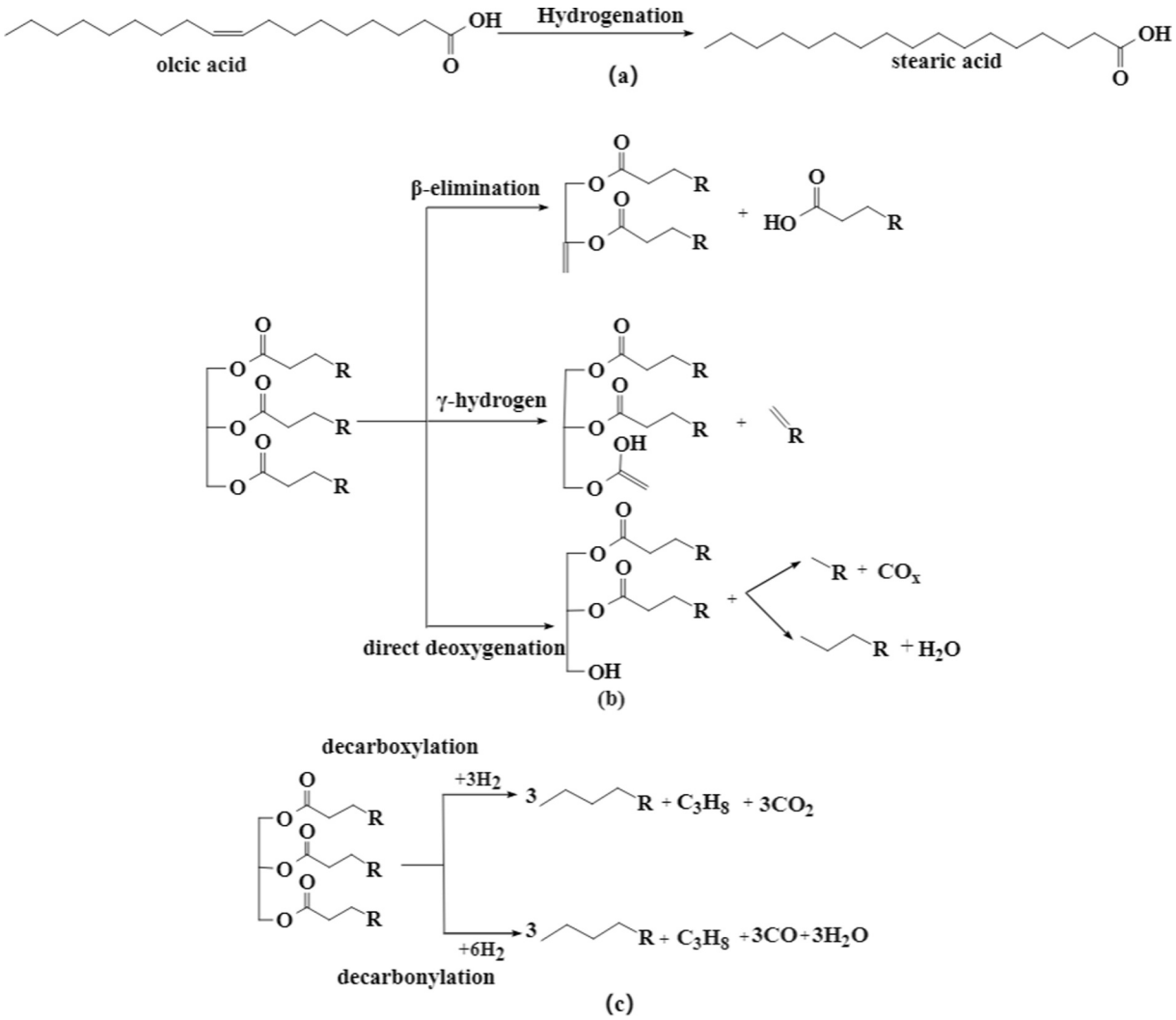


Fig. 4. Hydrodeoxygenation (HDO) mechanism. Hydrogenation of unsaturated fatty acids (a); Hydrogenation Conversion pathway of triglycerides (b) and decarboxylation pathway and decarbonylation (c).

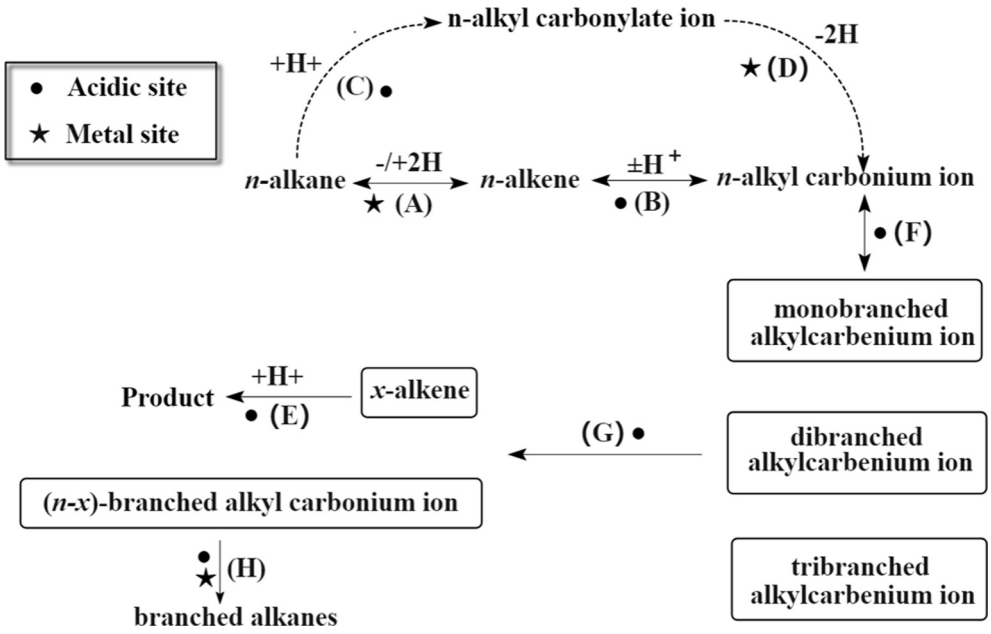


Fig. 5. Complete pathways for hydroisomerization and hydrocracking of n-Alkanes on bifunctional catalysts.

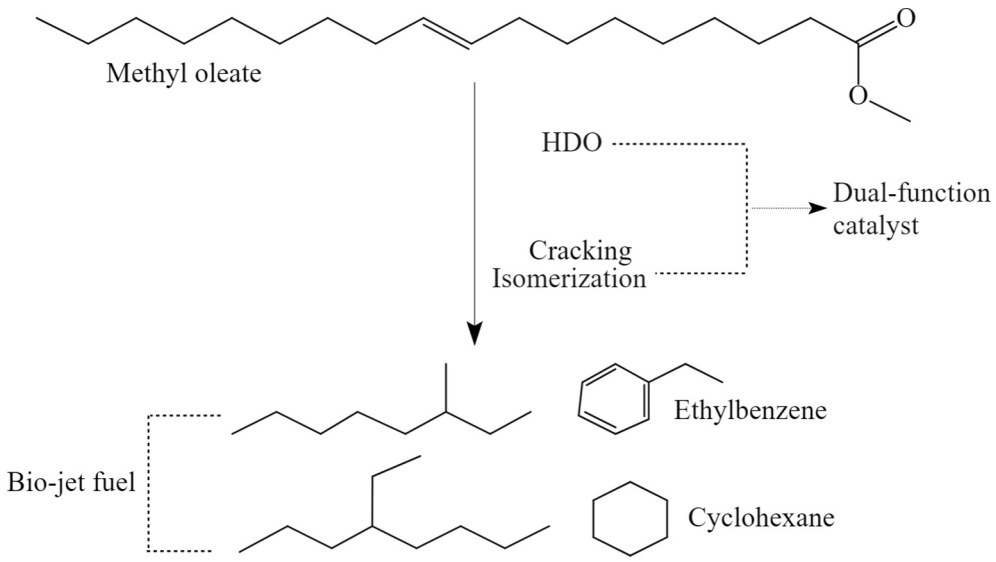


Fig. 6. One step preparation of bio-jet fuel.

pathway integrating two key steps: hydrothermal liquefaction (HTL) and hydrogenation upgrading. The HTL is the first stage of this process, primarily used to convert biomass into liquid "bio-crude oil"[41]. Hydrothermal liquefaction is not sensitive to the moisture and free fatty acid content of raw materials, and can handle almost any type of wet biomass, greatly expanding the source of raw materials, eliminating the need for expensive drying and pretreatment of WCO[42]. The core principle is that under medium temperature and high pressure (250 ~ 370°C, 5 ~ 20 MPa), using water as solvent, and making full use of the special physical and chemical properties of water in subcritical state, macromolecular polymeric organics in biomass are hydrolyzed, depolymerized, and repolymerized to form bio oil[43]. In addition to water solvent, methanol/ethanol could also be used as a solvent for the conversion of triglycerides to biofuels under supercritical conditions, and it is now widely used for bio-oil conversion[44,45].

The CHJ process primarily consists of four steps, as shown in Fig. 7. First, the raw materials undergo pretreatment. The pretreated oil enters the hydrothermal liquefaction reactor, where components such as triglycerides in waste cooking oil undergo hydrolysis, deoxygenation, decarboxylation, and polymerization reactions under the action of supercritical water and catalysts. These reactions convert the oil into lower-molecular-weight biocrude[41]. The biocrude is then pumped into a hydrogenation reactor where, under the action of a catalyst, it undergoes hydrogenation deoxygenation (HDO) upgrading treatment at high temperature, high pressure, and in a hydrogen atmosphere. Finally, through atmospheric or vacuum distillation, products such as naphtha, jet fuel fractions, and diesel are separated based on boiling point differences[19,46,47].

As an emerging technology, CHJ currently operates at the pilot and demonstration stages in terms of technical maturity, with few large-scale industrial applications. Its core advantage lies in significantly lower hydrogen consumption compared to HEFA. Unlike the HEFA process, CHJ achieves efficient deoxygenation through radical reaction mechanisms in a hydrothermal environment, resulting in higher aromatic content in the product and thereby enhancing fuel thermal stability. However, catalyst longevity faces severe challenges: the hydrothermal reaction environment (high-temperature, high-pressure water) demands exceptional hydrothermal stability from catalysts, with leaching of metal active components and collapse of the support structure being the primary causes of deactivation[48,49]. Table 2 shows the comparison of the analyses of HEFA and CHJ.

3. Application of different types of catalysts

3.1. Catalyst for hydrogenated of esters and fatty acids

The hydrogenation of esters and fatty acids comprises two steps. The first step involves hydrodeoxygenation(HDO), primarily aimed at reducing the O/C ratio and increasing the H/C ratio. Therefore, the catalyst design objective for this step is to achieve efficient C-O bond cleavage under mild reaction conditions while suppressing C–C bond cleavage and unnecessary aromatic ring saturation[16]. Due to catalyst

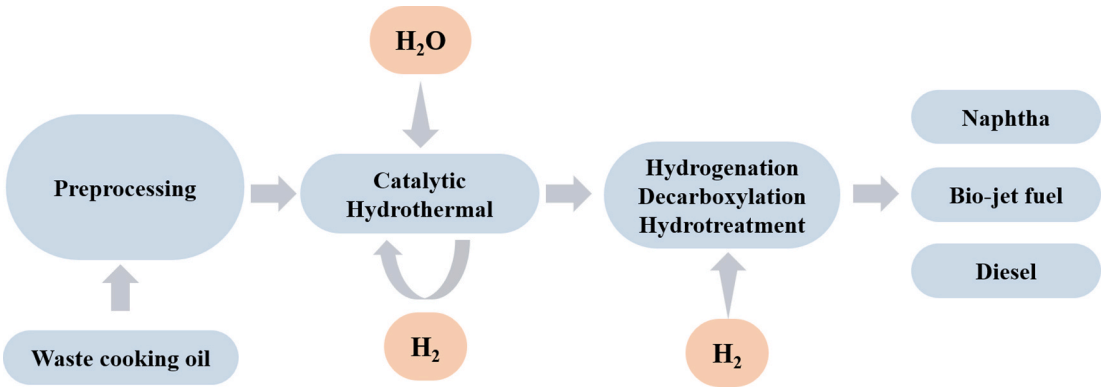


Fig. 7. Flow chart of catalytic hydrothermolysis jet.

Table 2
Comparison between CHJ and HEFA[16,32,47].

	Hydro Processed Esters and Fatty Acids (HEFA)	Catalytic Hydrothermolysis (CHJ)
Process steps	1.Hydrodeoxygenation2. Isomerization and catalytic cracking	1. Catalytic conjugation, cyclization & cross linking (preconditioning) 2. Cracking, hydrolysis, decarboxylation, isomerization, and cyclization (CHJ conversion) 3. Mild hydrotreating and fractionation (post refining)
Technology maturity	Commercialization	Demonstration Phase
Jet fuel standard	ASTM D7566 with 50% blending	ASTM D7566 with 50% blending
Liquid distillates	Naphtha, SAF and diesel	Gasoline, SAFand diesel
Feedstock	Vegetable oil, animal fat, or used cooking oil	Vegetable Oils, animal fats and waste greases.
Energy consumption	Hydrogen – High	Hydrogen – Low

deactivation, the generation of cracked species, and high process costs, the early catalysts developed for this step were noble metals loaded on zeolites or oxides, and later shifted to other transition metals such as Ni, Mo, Co or their loaded bimetallic composites [50,51].

The second step involves hydroisomerization and hydrocracking. The essence of isomerization-cracking catalysts lies in their dual-functionality, with performance highly dependent on the balance between metallic and acidic functions. This process selectively converts long-chain n-alkanes into high-value branched isomers and small-molecule hydrocarbons. The metal active sites and acidic sites within the isomerization-cracking catalyst exhibit synergistic effects. These metal sites typically employ precious metals such as platinum or palladium, while the acidic sites utilize supports like zeolite molecular sieves, modified alumina, and alumina[21,31]. The following is an analysis and comparison of different types of catalysts (Table 3).

3.1.1. *Bimetallic catalyst*

In recent years, the focus of catalyst research has shifted from single active metals to bimetallic systems. Their high performance in bio-oil HDO reactions stems from the synergistic regulation of dual-functional active sites through electronic effects, geometric effects, and strong metal-support interactions. This multiscale regulatory mechanism directly determines the catalyst's intrinsic HDO activity, selectivity for target hydrocarbon products, and long-term operational stability[52]. For example, the Ni-Mo bimetallic catalyst system has become a key choice among non-precious metal catalysts due to its unique synergistic effects. Its core advantage lies in achieving precise functional division and synergistic enhancement through the formation of a Ni-Mo-S mixed active phase: sulfur vacancies generated at Mo^{4+} sites, owing to their oxygen affinity, selectively adsorb and polarize C-O bonds in fatty acid molecules, laying the groundwork for deoxygenation reactions. Meanwhile, the Ni component acts as a highly efficient electron promoter,

Table 3
Analysis and comparison of different types of catalysts.

Catalyst type	Carrier/modification method	Active component	Performance highlights	literature
Ni-La/SiO_2	SiO_2	Ni-La	High dispersion, anti-sintering and anti-carbon deposition; efficient selective deoxygenation of fatty acid methyl esters.	[55]
Ni-Mo/SiO_2-TiO_2	Mesoporous SiO_2-TiO_2	Ni-Mo	High catalytic activity and selectivity in deoxidation of waste cooking oil.	[56]
Ni-Cu/SBA-15	SBA-15 (hexagonal mesoporous)	Ni-Cu	Compared with single metal Ni:	[60,62]
Ni-Mn/SBA-15		Ni-Mn	Lower light oil production rates and higher gas production rates; heavy oil fuel	
Ni-Zn/SBA-15		Ni-Zn	properties improved.	
Pt-Ni/CWMM- Co	Microwave activated coconut taro based activated carbon	Pt-Ni	The hydrocarbon yield was 33.1% (99.99% C_8-C_{16} selectivity), which was significantly better than that of single metal.	[58]
Ni-Co/SBA-15-SH	SBA-15(sulfhydryl functionalization)	Ni-Co	80% hydrocarbon yield, 92% aviation kerosene selectivity (C_8-C_{16}).	[62,63]
Ni-Co/SBA-15-NH_2	SBA-15(amino functionalization)	Ni-Co	83% C_8-C_{16} hydrocarbon yield, 88% aviation kerosene selectivity; Main product C_{15}-C_{17} alkane/olefin.	
Ni/Beta	Beta molecular sieve	Ni	The conversion of methyl laurate was 81%, and the selectivity of C_{11}-C_{12} alkanes was 46.5%.	[64]
Pd@Al_x-mSiO_2	Al Doped Mesoporous SiO ɑ shell	Pd	Methyl palmitate conversion 98%, alkane selectivity > 99%.	[65]
Ni-Co/TiO_2	TiO_2	Ni-Co	High deoxygenation conversion of 78.78% was achieved and excellent selectivity for C_{10}-C_{16}(Meets Bio-jet fuel standards) hydrocarbons was demonstrated.	[79]
Pd/multistage pore ZSM-5	Multistage hole ZSM-5	Pd	The dodecane formation efficiency was 8.5 times that of the traditional catalyst; Acid sites and PD space isolation inhibit side effects.	[34]
Ni/Mesoporous Y	Mesoporous Y	Ni	Mesoporous, high surface area; Bio-jet fuel yield reached 58.4%.	[80]
Mo/H-ZSM-5	H-ZSM-5	Mo	3D 10-MR cross-channels, strong acid; 83.6% Bio-jet fuel selectivity.	[81]
Ca-Ni/ZSM-5	Z_1M-5	Ca-Ni	Volcanic mud + marble waste, Ca inhibits cracking; C_{15}-C_{17} selectivity 66.6%.	[72]
Pt/SAPO-11	SAPO-11	Pt	One-dimensional 10-membered ring pore structure, tunable medium acidity; the hexadecane coverage rate reached 81.8%, isomerism −selectivity rate reached 86.5%.	[66]
Ni-$H_3PW_{12}O_{40}$/HAP	$H_3PW_{12}O_{40}$	Ni	High-temperature and high-pressure resistance, stability and durability, not easy to accumulate carbon; Liquid phase product yield over 80%, with 54–77% selectivity for Bio-jet fuel	[82]
PO_4-ZrO_2	H_3PO_4	ZrO_2	P-OH/Zr^{4+} enhances Brønsted/Lewis acidity; bio-jet selectivity 33.92%, liquid yield 39.44%.	[74]
Pt/SAPO-11	SAPO-11	Pt	n-Hexadecane conversion was 81.8% with a selectivity of 86.5%	[66]
Ni/ZSM-5	ZSM-5	Ni	Exhibits excellent hydrogenation catalytic performance	[67]
Ni/Beta	Beta	Ni	280°C, 1 h; STY: 33.4 $mmol \cdot g^{-1} \cdot h^{-1}$ (C_{11}-C_{12}) (Meets Bio-jet fuel standards)	[83]
Pt-Sn/M−SAPO−11	SAPO-11	Pt-Sn	Bio-jet fuel yield > 60%; Freezing point ≤ -40°C; Optimized branching	[68]
NiMo/USY@Al-SBA-15	USY@Al-SBA-15	NiMo	WCO conversion of 90–100% to 39.7% jet fuel selectivity, Isoparaffin/normal alkane ratio up to 2.	[84]
NiCo/TiO_2	TiO_2	NiCo	Deoxygenation: 83.13%; Bio-jet selectivity: 44.5%; Meets ASTM standards	[85]
$CoAl_2O_4$	Al_2O_4 weak Brønsted acid	Co	Over 90% WCO conversion rate, Bio-jet fuel (C_8-C_{16}) selectivity: 20%	[86]

significantly enhancing H_2 heterolytic dissociation capability and providing abundant hydrogen overflow species. This synergy between electronic and geometric effects endows it with higher intrinsic activity on the hydrogenation-deoxygenation pathway compared to single-metal catalysts[53,54].

Assessing the relative advantages and industrial potential of different catalyst systems requires in-depth investigation into their structure–activity relationships, as the advantages and performance trade-offs resulting from different metal combinations vary significantly. Jiaqi Yan et al. prepared Ni-La/SiO_2 catalysts via the coprecipitation method. Within the system, La^{3+} induces strong metal-support interactions, acting as a "structural aid" to stabilize highly dispersed Ni nanoparticles and effectively suppress their sintering during high-temperature reactions. This enhanced physical stability directly correlates with the catalyst's sustained deoxygenation activity, enabling the efficient and highly selective conversion of fatty acid methyl esters into long-chain alkanes[55]. Similarly, the NiMo/SiO_2-TiO_2 catalyst developed by Lin Fu et al. demonstrates dual synergistic effects between the support and metal: the mesoporous SiO_2-TiO_2 composite support not only provides tunable acidic sites, but its TiO_2 component further promotes the dispersion and stability of metal nanoparticles through strong metal-support interactions (SMSI). Simultaneously, the introduction of Mo species optimizes the active sites' adsorption and polarization capabilities for C=O/C-O bonds in oxygen-containing intermediates. Synergizing with Ni's hydrogenation function, this efficiently directs the reaction pathway toward hydrogenation-deoxygenation, yielding high-yield renewable alkanes under mild conditions[56]. A composite carrier was prepared by doping sulfur-resistant SiO_2 with TiO_2 exhibiting excellent resistance to metal poisoning and chlorine attack, resulting in a robust anti-poisoning catalyst system. In the same way, the ZrO_2/TiO_2 composite oxide carrier demonstrated outstanding sulfur tolerance and regeneration performance during the catalytic process[57]. In the precious-metal/non-precious-metal system, the Pt-Ni/CWMM-CO catalyst synthesized by Fenti Visiamah et al. achieved a 33.1% aviation kerosene hydrocarbon yield (with 99.99% selectivity for C_8–C_{16} hydrocarbons) under hydrocracking conditions at 550°C [58]. Research indicates that bimetallic catalysts not only enhance hydrogenation activity but also generate additional acidic sites at the metal-support interface, thereby simultaneously increasing both the rate of aromatic hydrogenation and the rate of oxygen removal[59].

Shinyoung Oh et al. prepared three nickel-based bimetallic catalysts on hexagonally structured mesoporous SBA-15 carriers (NiCu/ SBA-15, NiMn/SBA-15, and NiZn/SBA-15) and comparatively analyzed the effects of single and bimetallic on bio-oil hydrodeoxygenation activity. Compared to the single-metal Ni/SBA-15 catalyst, the bimetallic design significantly alters the properties of Ni active sites through electronic effects, generally reducing light oil yield while substantially increasing the proportion of gas products[60,61].

In addition to adjusting the intrinsic composition of bimetallic systems, surface functionalization of the support is another effective strategy for regulating the microenvironment of active sites. For example, the research by Noor Azira Abdul Razak et al. provides a typical case: they functionalized SBA-15 mesoporous molecular sieves with sulfhydryl (−SH) groups and then loaded them with a Ni-Co bimetallic compound. The core of this design lies in the −SH groups acting as strong anchoring sites, effectively enhancing the dispersion of the metal precursor and suppressing its sintering during reduction and reaction processes. Characterization confirmed that the obtained Ni-Co/SBA-15-SH catalyst exhibits higher metal dispersion and tunable acidity while maintaining its hexagonal mesoporous structure. Consequently, this catalyst demonstrated significantly optimized performance in the hydrogenation deoxygenation of WCO, achieving not only an overall hydrocarbon yield of 80% but, more critically, a selectivity of up to 92% for the target jet fuel fraction (C_8-C_{16})[62]. Complementary to this, Darfizzi Deraw et al prepared novel bimetallic NiCo/SBA-15-NH_2 catalysts by loading nickel and cobalt onto amino-functionalized SBA-15 mesoporous carriers by the wet impregnation method, and the characterization showed that the catalysts maintained a hexagonal mesoporous structure (specific surface area 393.93 m^2/g), the metal particles were uniformly dispersed and formed weak/strong acid synergistic sites. The introduction of –NH_2 to enhance the Lewis acid sites promoted decarboxylation and C–C bond cleavage, and 83% C_8-C_{20} hydrocarbon yield and 88% jet fuel selectivity were achieved under optimized conditions (330°C, 5% catalyst loading, 300 min), with the main products being C_{15}-C_{17} alkanes/olefins. The above two Ni-Co bimetallic systems (Ni-Co/SBA-15-SH) synergistically enhance the deoxygenation efficiency through electronic and geometrical effects, and the sulfur-modified (−SH) SBA-15 carrier enhances the metal dispersion and inhibits the carbon accumulation. The SBA-15 carrier modified by an amino group (NH_2) enhances the dispersion of Ni-Co alloy (particle size of 5 nm) and inhibits excessive cracking[63].

Bimetallic catalysts, through multi-metal synergistic effects and support engineering, enhance hydrogenation deoxygenation activity and selectivity while also facing inherent trade-offs such as product distribution regulation (e.g., excessive cracking)[60]. Its industrial potential depends not only on intrinsic activity but also on resistance to poisoning, long-term stability, and the economic balance of non-precious metal substitutions[56]. Therefore, rational catalyst design requires synergistic optimization of the active site microenvironment and cost-effectiveness to drive practical applications.

3.1.2. *Bifunctional catalysts*

Beyond bimetallic catalysts, the precise "spatial configuration design" of metal sites and acidic sites represents another cutting-edge strategy for achieving highly efficient and selective synergistic catalysis. For example, Tinghu Wang et al prepared Ni/Beta bifunctional catalysts for the preparation of bio-aviation coal from the hydrodeoxygenation of methyl lauric acid to achieve the highest conversion (81%) and C_{11}-C_{12} alkane selectivity (46.5%)[64]. Xincheng Cao et al reported a stable and efficient HDO catalyst (Pd@Alx-mSiO_2) with a uniformly dispersed palladium (Pd) core inside and a mesoporous silica shell doped with aluminum (Al) atoms on the outside. Al doping was introduced into the Lewis and Brønsted acid sites, which synergistically acted with Pd, and resulted in a methyl palmitate conversion of up to 98% and the alkane selectivity was > 99%[65]. Moreover, mesoporous silica in catalysts combines the advantages of SiO_2—resistance to sulfur, chlorine, and metal poisoning—with its own inherent strengths, making it one of the carriers with the strongest overall resistance to impurities currently available. Shengzhe Ding et al reported a Pd-doped multistage-pore ZSM-5 catalyst with Pd nanoparticles selectively loaded in mesopores and acid sites located in micropores, which were spatially isolated to avoid side reactions. In lauric acid hydrodeoxygenation, the dodecane generation efficiency was 8.5 times higher than that of conventional catalysts[34]. The fundamental purpose of these catalysts is to maximize the efficiency of the main reaction by precisely controlling the molecular transport pathways and reaction sequences of the reactants. This marks a shift in catalyst design from traditional "material screening" to a new phase of "system construction", providing a core approach for developing next-generation customized catalysts for high-performance biorefineries in the future.

Bifunctional catalysts integrate metallic sites (responsible for hydrogenation/dehydrogenation) with acidic sites (responsible for isomerization, cracking, and dehydration), enabling them to catalyze multiple sequential reactions in series. This dual functionality is pivotal for achieving a one-step hydrogenation-deoxygenation-isomerization process for bio-oils. The core of their design lies in the precise matching and synergistic interaction between metallic and acidic functionalities. For example, the classic Pt/SAPO-11[66] and Ni/ZSM-5[67] catalysts utilize precious metals or non-precious metals to provide hydrogenation activity, respectively, and rely on the shape-selective acidic channels of molecular sieves to direct isomerization reactions[67]. Such catalysts combine hydrogenation activity with isomerization capability, enabling

the simultaneous completion of deoxygenation and isomerization reactions in a single step. Additionally, Zheng Zhifeng's team at Xiamen University developed a Pt-Sn/M−SAPO−11 catalyst. This catalyst features Pt_xSn_γ alloy loaded onto modified SAPO-11 molecular sieves (M = Zr, Ti) for the one-step catalytic reaction of fatty acid methyl esters[68]. The mild acidity of SAPO-11 (weak Brønsted acid) suppresses excessive cracking, while the Pt-Sn alloy enhances dehydrogenation-hydrogenation activity. This yields aerosol products with $>$ 60% yield and a freezing point $\leq$ -40°C, significantly optimizing the branched structure of the final product.

Breakthroughs in bifunctional catalysts performance rely on comprehensive and precise control spanning from the nanoscale (metal particle dispersion) to the molecular scale (acid strength and type), meticulously balancing deoxygenation and isomerization pathways to directly and efficiently produce Bio-jet fuel meeting specifications. Its design philosophy has shifted from screening to construction, relying on precise spatial configuration control to synergistically enhance performance [65,68]. However, its high performance is constrained by the delicate balance of active sites and the complexity of preparation. Scalable application must address challenges including resistance to poisoning, cost control, and structural reproducibility.

3.1.3. Molecular sieve catalysts

The realization of high-performance catalysts relies not only on the optimization of active metal sites (such as bimetallic synergism) but also on the precise design of synergistic "acid-functional" carriers. In recent years, molecular sieves have received wide attention and applications as inorganic crystalline materials (e.g., zeolite molecular sieves, aluminum phosphate molecular sieves, etc.) or organic framework materials (e.g., MOFs, COFs) with regular and uniform pore structure (usually microporous, $<$ 2 nm), as well as a bright application prospect in the field of catalysis. Zeolite molecular sieves are crystals composed of silicates or aluminosilicates, whose oxide-based framework consists of tetrahedra sharing corners, where T denotes the tetrahedral atom, most commonly Si and Al. Different tetrahedral connectivity patterns result in distinct zeolite framework types[69]. This material exhibits unique shape selectivity, high thermal stability, and strong acidity. Its molecular structure promotes the diffusion of reactants and products, making it suitable for hydrocracking and hydroisomerization reactions[70].

To date, 235 distinct zeolite framework types have been identified in natural or synthetic zeolites. Brønsted sites and Lewis sites are the two most common active sites in zeolite-mediated material transformation reactions[40], e.g. The reaction mechanism for biomass conversion at Brønsted acid sites involves bridging hydroxyl groups (Si-OH-Al) within the zeolite framework providing protons (H^+) to catalyze dehydration, condensation, and alkylation reactions. Electron-deficient metal centers within the framework (e.g., Sn^{4+} and Ti^{4+}) catalyze redox reactions and molecular rearrangements by accepting electron pairs, representing biomass conversion reactions at Lewis acid sites.

Among numerous zeolite molecular sieves, H-ZSM-5 is a zeolite material featuring a three-dimensional decagonal ring cross-linked pore structure. Its interior contains two intersecting pore systems with a stable framework. Due to its strong acidity, direct application in the conversion of long-chain alkanes often leads to excessive cracking. Research efforts therefore employ various modification strategies to balance its dehydrogenation activity with selectivity toward target products. For example, Indriana Kartini et al. demonstrated that Mo-loaded H-ZSM-5 can be used for the dehydrogenation of palm kernel oil. Its acidic sites synergize with metal sites to yield highly selective C_8–C_{16} hydrocarbon fractions[71]. Hartati et al. synthesized a Ca/ZSM-5 support using volcanic ash and marble waste, then loaded Ni to form a bifunctional catalyst. The introduced Ca^{2+} provided basic sites that effectively neutralized some strong acid centers, suppressing cracking side reactions. This significantly enhanced the selectivity of C_{15}–C_{17} long-chain hydrocarbons in the deoxygenation products of waste cooking oil by 18.9% compared to unmodified ZSM-5, reaching 66.6%[72]. This directly demonstrates that altering the acidity of the carrier through chemical modification is an effective means of controlling product distribution.

SAPO-11 (AEL) is a prominent member of the aluminium silicate-phosphate molecular sieve family. With its unique one-dimensional decahydrocyclohexene ring structure and adjustable moderate acidity, it holds an almost irreplaceable position in the selective isomerization of long-chain n-alkanes, minimizing cracking side reactions to the greatest extent. SAPO catalysts for alkane hydroisomerization are usually supplemented with Pd and Pt to obtain high activity and selectivity[73]. For example, the SAPO-11 molecular sieve prepared by Yang, Sanni et al. achieved a conversion rate of up to 81.8% for n-hexadecane and an isomerization selectivity of 86.5% under optimized conditions[66]. This study clearly demonstrates the high efficiency of the tandem catalytic cycle formed by Pt sites (dehydrogenation/hydrogenation) and SAPO-11 acid sites (isomerization).

In developing bifunctional catalysts for the hydrogenation-deoxygenation and isomerization of bio-oils, the rational selection and modification of acidic supports are crucial. The core design principle involves selecting a fundamental carrier based on the target reaction. Through methods such as metal loading, ion exchange, heteroatom doping, or structural engineering, the type, strength, and distribution of the carrier's acidity are precisely controlled to match the characteristics of the metal active sites. This enables the efficient and highly selective conversion of bio-oils into target fuel components. In industrial applications, molecular sieves exhibit excellent thermal stability. However, achieving precise control over their acid distribution and pore structure at scale is costly. It is also sensitive to impurities in raw materials (H_2O, S), and its applicability depends on the balance between its resistance to poisoning and the economic viability of modification.

3.1.4. Other catalysts

Catalyst design is no longer confined to traditional molecular sieves; other materials with tunable acidity or ordered structures are also being extensively explored. The phosphate-modified nano-zirconia catalyst reported by Saviola et al. achieved a Bio-jet fuel selectivity of 33.92% in the hydrogenation conversion of waste palm oil by forming synergistic acidic sites comprising P–OH (Brønsted acid) and Zr^{4+} (Lewis acid)[74]. Metal-organic framework-derived materials (MOFs) or carbon-based composites derived from them as precursors demonstrate immense potential in biomass conversion reactions such as furfural hydrogenation and triglyceride hydrogenation. Their ultra-high specific surface area and customizable pore/metal site architecture offer novel insights for designing novel bifunctional catalysts[75–77]. Carbon-based materials, with their exceptional chemical inertness, corrosion resistance, hydrophobicity, and highly tunable pore structure, effectively resist chemical erosion from sulfur and chlorine, mitigate the effects of alkali metals, and buffer carbon deposits. They have been experimentally proven to be a highly stable carrier choice for treating WCO[78].

3.2. Catalyst for catalytic Hydrothermolysis jet

The CHJ process comprises two distinct catalytic stages: initial Hydrothermal Liquefaction (HTL) and subsequent Hydrodeoxygenation (HDO) upgrading[87]. The catalyst requirements and challenges differ significantly between these stages and from those in the HEFA process, particularly concerning the harsh hydrothermal environment.

3.2.1. Catalysts for the hydrothermal liquefaction (HTL) stage

Under specific conditions, the HTL process can proceed without a catalyst. However, catalysts can significantly enhance reaction rates, increase biological yield, and reduce oxygen content under milder conditions[88]. The catalyst used in this stage must be capable of withstanding high-temperature, high-pressure water[89]. Alkali catalysts (e.g., K_2CO_3, Na_2CO_3, KOH) are widely used. They promote the hydrolysis of triglycerides into fatty acids and glycerol, and facilitate

decarboxylation/dehydrogenation reactions, thereby increasing biocrude yield and reducing its viscosity and oxygen content[90]. Organic acids (e.g., formic, acetic acid) can act as in-situ hydrogen donors, aiding stabilization of reactive intermediates. Solid catalysts offer easier separation and potential reuse. Common types include: Zeolites (e.g., H-ZSM-5), metal oxides (e.g., CaO, ZrO_2). They catalyze cracking, dehydration, and ketonization reactions. Additionally, Redox-active metal oxides(Fe_3O_4, CeO_2)can catalyze redox reactions, facilitating the removal of oxygen as CO or CO_2. The choice of HTL catalyst depends on the feedstock and desired biocrude properties. A key trade-off is that while catalysts can improve biocrude quality, they may also suffer from deactivation due to coking or metal leaching in the aggressive hydrothermal medium.

3.2.2. Catalysts for the hydrodeoxygenation (HDO) upgrading stage

The biocrude from HTL requires further HDO to produce specification-compliant jet fuel. The catalyst employed here is conceptually similar to the HDO step in the HEFA process, requiring a combination of metal sites (for hydrogenation) and acid sites (for dehydration and cracking) [91]. However, the feedstock (biocrude) is more complex than pure triglycerides, containing a wider range of oxygenates, unsaturated compounds, and often higher nitrogen and sulfur content. The appropriate catalyst should be selected based on the feedstock characteristics and process objectives.

For traditional HDO catalysts, Transition metal sulfide catalysts (e.g., NiMoS/γ-Al_2O_3, CoMoS/γ-Al_2O_3) are effective for the simultaneous removal of oxygen, sulfur, and nitrogen. Noble metals (Pt, Pd) on various supports are also active but more sensitive to impurities[92]. Research is exploring tailored catalysts for biocrude upgrading. For instance, iron-based catalysts (Fe_2O_3/γ-Al_2O_3) have shown promise due to their redox properties, promoting decarboxylation pathways[47]. Nickel supported on ordered mesoporous silica (e.g., Ni/SBA-15) has been effective in the catalytic pyrolysis and deoxygenation of oils, producing hydrocarbons in the jet fuel range[93].

3.3. Catalyst challenges from an industrial application perspective

Although laboratory studies demonstrate that various catalysts exhibit outstanding performance in WCO conversion, their industrial application necessitates systematic evaluation of scalability, economic viability, and long-term robustness under real-world conditions[94]. Catalyst mass production involves complex engineering scaling challenges, such as ensuring uniform distribution of active components across macroscopic carriers, guaranteeing mechanical strength, and optimizing forming processes. These factors collectively determine the final cost and performance of the catalyst[95].

In terms of economics, while non-precious metal catalysts (such as Ni, Co, Mo) offer advantages in raw material costs, they typically require more stringent operating conditions. Additionally, issues like coking and sintering may necessitate more frequent regeneration or shorten service life, thereby impacting total lifecycle costs. Precious metal catalysts, though highly active, are expensive and more sensitive to impurities like sulfur and chlorine in feedstocks, limiting their application scenarios. Beyond this, the economic viability of the process also heavily depends on the availability and cost of external hydrogen sources. The current mainstream "gray hydrogen" significantly undermines the lifecycle carbon reduction benefits of WCO-SAF, while the use of "green hydrogen" or "blue hydrogen" is an inevitable choice for achieving deep decarbonization. However, this also poses challenges for project siting and cost structure[96].Comparing the two mainstream routes, HEFA and CHJ, reveals significant differences in their technical and economic viability. The HEFA process is mature and currently dominates production capacity, but its high hydrogen consumption and stringent raw material quality requirements pose major constraints. The CHJ process offers potential advantages in feedstock adaptability (capable of processing high-moisture, high-acid-value materials) and hydrogen consumption, making it more suitable for treating low-quality waste crude oil. However, its high-pressure reaction environment imposes higher demands on equipment engineering and the hydrothermal stability of catalysts.

Catalyst deactivation is a core challenge in industrial operations. Unlike refined vegetable oils, waste cooking oils contain complex impurities originating from the cooking process, food residues, and storage contamination. These primarily include sulfur, chlorine, and phosphorus compounds, along with trace heavy metals[97]. Such impurities irreversibly poison the active sites of metals through chemical adsorption or react with the acidic centers of the carrier, permanently impairing the catalyst's hydrogenation and acid functions. Therefore, the development of a catalyst system that combines high activity with strong resistance to poisoning.

Carbon-based materials (activated carbon, carbon nanotubes) leverage their chemical inertness, hydrophobicity, and high specific surface area to primarily accommodate impurities through physical adsorption[98]. They provide effective buffering for active metals, demonstrating outstanding comprehensive resistance. Aluminum phosphate molecular sieves (e.g., SAPO-11) inherently tolerate phosphorus impurities due to phosphorus-containing frameworks, while their moderate acidity helps maintain isomerization activity and stability. In contrast, conventional γ-Al_2O_3 readily deactivates upon phosphorus binding, and the micropores of silica-alumina zeolites are prone to impurity blockage[99]. When preparing catalysts for WCO, resistance to poisoning must be prioritized as equally important as intrinsic activity. This requires combining tolerant supports (e.g., carbon materials, SAPOs) with robust active phases (e.g., NiMo, Pt-Sn alloys)[100]. Furthermore, constructing composite oxides (e.g., TiO_2-ZrO_2) or designing functionalized supports with core–shell structures and hierarchical pore systems to synergistically build multi-layered protection at both physical and chemical levels represents a key direction for developing next-generation highly stable WCO treatment catalysts[101].

In summary, future R&D efforts must shift from a purely activity-driven approach to comprehensive performance and cost optimization across the entire lifecycle. Emphasis should be placed on conducting long-term operational testing using real, fluctuating WCO feedstock at pilot or industrial demonstration scales. This should be coupled with technical–economic analysis and life cycle assessment of the system to accelerate the commercialization of high-performance catalysts.

4. Influence of process conditions and optimization

The efficiency and economics of bio-aviation fuel preparation fundamentally depend on precise control of core operational parameters. These parameters not only directly influence reaction rates and thermodynamic equilibrium but also profoundly regulate reaction networks and the intrinsic properties of catalysts through complex interactions.

4.1. Effect of reaction temperature and pressure

Temperature and pressure, as the most fundamental thermodynamic variables, must be set in close alignment with the chemical nature of the target reaction stage. During the initial hydrogenation and saturation phase of the HEFA process, the primary objective is to eliminate the double bonds in the glycerol triglycerides present in the feedstock. This exothermic reaction ($\Delta H < 0$) is thermodynamically more favorable at lower temperatures (200–300°C) and effectively suppresses side reactions such as C–C bond cleavage. For example, Sinopec's SRJET technology employs a Ni-based catalyst at 250°C and 5 MPa to convert triglycerides into fatty acids via β-hydrogen elimination during the hydrogenation saturation stage [94]. However, low temperatures imply slower reaction kinetics, which must be compensated for by extending residence time or enhancing catalyst activity. The accompanying hydrogen pressure (typically 3–6 MPa) must be sufficient to ensure adequate hydrogen coverage on the catalyst surface, thereby achieving

high conversion rates[102].

Entering the core stage of hydrogenation deoxidation (HDO), the effects of temperature and pressure become more complex. Temperature is the primary factor determining the selectivity of the reaction pathway. Studies indicate that the three primary dehydrogenation pathways, hydrogenation (HDO), decarboxylation (DCO), and decarboxylation oxidation (DCOx), exhibit distinct activation energies. Moderately elevated temperatures (350–400°C) significantly accelerate all pathways, ensuring complete conversion of feedstocks. For instance, waste lard achieves a 96.62% conversion rate at 400°C [103]. However, the effect of temperature on selectivity is nonlinear: below a certain threshold (e.g., 300°C for specific catalysts), it may predominantly promote the HDO pathway, yielding straight-chain alkanes matching the carbon number of the feedstock. Beyond this threshold, pathways causing carbon loss (e.g., DCO/DCO_x) and secondary reactions like cracking and aromatization intensify[104,105]. Dynamic studies by Yuwei Bie et al. indicate that for HDO of fatty acid methyl esters, optimal equilibrium can be achieved at 300°C and 8 MPa. Reaction temperature significantly influences the HDO reaction pathway of methyl esters, while hydrogen pressure exerts a relatively minor effect[104]. Compared to the dominant role of temperature, hydrogen pressure exerts a relatively weaker direct influence on pathway selectivity. Its primary function lies in providing an ample hydrogen supply for the hydrogenation step by enhancing the solubility of liquid-phase hydrogen and increasing the partial pressure of hydrogen at the catalyst surface. This indirectly promotes the HDO pathway and rapidly saturates unsaturated intermediates, thereby inhibiting their polymerization and coking. Consequently, it maintains catalyst activity and stability.

The temperatures in the isomerization and cracking stages are typically lower than or equal to those in the HDO stage, with a typical operating temperature range between 340°C and 400°C. Temperature is the key parameter for controlling cracking depth and isomerization selectivity. Isomerization reactions generally dominate at relatively lower temperatures. As temperature increases, the rate of hydrocracking reactions significantly rises, leading to the breakage of long-chain alkanes. Moderately increasing temperature can enhance the yield of jet fuel fractions (C_8-C_{16}). However, excessively high temperatures cause excessive cracking, generating large amounts of light gases (C_1-C_4) and gasoline fractions, resulting in a sharp decline in jet fuel yield. For example, on Pt/Beta catalysts, an increase in reaction temperature from 300°C to 400°C results in a decrease in liquid product yields and an increase in gas and coke yields, while aromatics yields increase from 4.5 to 15.63%[106]. The elevated hydrogen pressure at this stage suppresses the polymerization and coking of olefins generated by dehydrogenation reactions at the acidic sites of the catalyst, thereby protecting the catalyst and extending its operational lifespan. Simultaneously, the high-pressure environment facilitates rapid hydrogenation of unsaturated intermediates, enhancing product stability. Typical pressure ranges reported in the literature are 7–10 MPa.

Compared to the more mature HEFA process with distinct reaction steps, one-step hydrogenation and catalytic hydrothermal cracking processes couple more complex reaction networks within the reactor, making them more demanding in terms of temperature and pressure requirements. To achieve efficient deoxygenation, cracking, and isomerization simultaneously within a single reaction space, one-step processes typically require precise balancing within a relatively narrow operating window[107]. The optimal deoxidation reaction parameters typically fall within the range of 350–400°C and 1–5 MPa[108]. However, the specific optimal value strongly depends on the catalyst system and feedstock characteristics. For example, ZSM-22 molecular sieves have the best isomerization efficiency at 320°C (72% of isoparaffins), but exceeding 350°C leads to molecular sieve skeleton collapse [109,110].

In contrast, the “temperature-pressure” synergistic regulation in the catalytic hydrothermal decomposition (CHJ) process exhibits a two-stage characteristic and generally requires higher pressure. In the first stage of hydrothermal liquefaction (HTL), the core process involves converting wet biomass into bio-crude oil using subcritical water. This procedure typically occurs in a medium-temperature, high-pressure aqueous environment, with typical operating temperatures ranging from 250–380°C. Pressure must be maintained at elevated levels of 5–30 MPa to sustain water in its liquid state and facilitate reactions such as hydrolysis and pyrolysis[111]. Temperature is the decisive factor for bio-crude oil yield at this stage, with a clearly defined optimum value. For example, using microalgae with 78.4% water content as feedstock, a bio-crude oil yield of 37% can be achieved at 300°C and 10 MPa[112]. BAO et al. found that at 310°C, the bio-crude oil yield reached a peak of 36.75%[113]. Research indicates that the energy provided by hydrothermal liquefaction reactions at low temperatures is insufficient, resulting in low conversion rates. As temperatures rise, the dissolution rate of organic compounds accelerates, promoting multiple reactions (hydrolysis, decarboxylation, aromatization, etc.), thereby accelerating the formation of bio-crude oil. However, excessively high temperatures cause intermediate products in this process to readily form coke and increase gas yield[114]. The subsequent bio-crude hydrogenation upgrading stage operates on a similar principle to HEFA hydrogenation. However, since its feedstock (bio-crude) typically contains a higher proportion of unsaturated bonds and aromatic structures, it often requires higher hydrogen partial pressures (e.g., 10–20 MPa) to promote deep hydrogenation saturation and suppress condensation coking reactions. Consequently, the overall pressure requirements of the CHJ process are significantly higher than those of HEFA and one-step processes using oils as feedstock.

The effects of temperature and pressure are not isolated but act synergistically on reaction equilibrium and rate. When increasing temperature to pursue high conversion rates, hydrogen pressure must be raised simultaneously to suppress the heightened risk of coking and deactivation. The synergistic requirements for temperature and pressure in different Bio-jet fuel preparation processes fundamentally reflect the characteristics of the reactants, differences in catalysts (such as the varying hydrogenolysis activities of noble metals versus transition metals), and the degree of coupling with the target reaction pathway. These parameters must be jointly determined based on the interplay between "catalyst properties, feedstock characteristics, and target product distribution"[115].

4.2. *Effect of Hydrogen-to-Oil ratio*

The impact of the hydrogen-to-oil ratio (defined as the ratio of hydrogen gas volume to feedstock oil volume under standard conditions, H_2/oil) is multidimensional. The impact of hydrogen-to-oil ratio on the hydrogenation esterification and fatty acid (HEFA) process is primarily reflected in three aspects: reaction pathways, product selectivity, and catalyst lifetime. A higher hydrogen-to-oil ratio implies a higher hydrogen partial pressure, which promotes the hydrogen-consuming HDO pathway, enhances the selectivity of straight-chain alkanes, and effectively delays catalyst deactivation by carbon deposits through surface flushing[103]. Conversely, low hydrogen oil ratios (e.g., <500 Nm^3/m^3) restrict hydrogen surface coverage. This not only shifts the reaction toward low-hydrogen-consumption DCO/DCOx pathways, leading to reduced jet fuel selectivity (potentially below 35%), but also accelerates catalyst deactivation due to intermediate polymerization [116,117].

Research indicates that under startup conditions characterized by low oil-to-hydrogen ratios and low hydrogen pressure, carbonaceous deposits permanently adsorb onto active sites, leading to severe and irreversible catalyst deactivation[117]. Therefore, during the hydrogenation saturation and HDO stages, a higher hydrogen-to-oil ratio (typically recommended at 500–1200 Nm^3/m^3) is essential to ensure reaction depth and catalyst stability. However, in the isomerization stage, excessively high hydrogen partial pressure may overly suppress the

dehydrogenation step at metal sites (a prerequisite for generating olefin intermediates for subsequent acid-catalyzed isomerization). Thus, the hydrogen-to-oil ratio must be moderately reduced in this stage (e.g., 300–1000 Nm^3/m^3) to maintain the delicate equilibrium between metal sites (dehydrogenation/hydrogenation) and acidic sites (isomerization) within the bifunctional catalyst[107].

The one-step process also relies on external hydrogen, but its optimization objectives are more complex: excessively high hydrogen partial pressure inhibits the dehydrogenation step that generates olefin intermediates at metal sites, a prerequisite for subsequent acid-catalyzed isomerization. Therefore, the one-step process must balance promoting hydrogenation-deoxygenation with maintaining essential dehydrogenation activity. For example, when using SDBS-Pt/SAPO-11 to catalyze the oxidation of castor oil (Jatropha oil), the activity is approximately 1000 Nm^3/m^3[118]. For the CHJ process, the hydrogen concept in its first stage has been fundamentally redefined. The reaction relies on in-situ hydrogen production via aqueous-phase reforming, making the water-to-oil ratio—rather than the hydrogen-to-oil ratio—the core parameter. The water-to-oil ratio not only influences the physicochemical properties of the reaction medium but also directly correlates with the in-situ hydrogen production rate. In the subsequent bio-crude hydrogenation upgrading stage, although external hydrogen supplementation is required, the objective is to achieve deep saturation of highly unsaturated complex molecules. Consequently, the required hydrogen partial pressure is typically far higher than in the HEFA process (reaching 10–20 MPa).

The hydrogen-oil ratio fundamentally represents the macroscopic manifestation of hydrogen chemical potential within the reaction system. A high hydrogen-oil ratio (high hydrogen partial pressure) unequivocally benefits all hydrogenation steps by promoting the HDO pathway, enhancing alkane selectivity, and effectively delaying catalyst deactivation due to coking. However, a delicate trade-off exists during the isomerization stage: excessively high hydrogen partial pressure inhibits the formation of olefin intermediates on metal active sites, which are essential precursors for acidic sites to perform skeletal isomerization. Therefore, the industrial practice often employs a staged hydrogen-to-oil ratio adjustment strategy: a high hydrogen-to-oil ratio is maintained in the main hydrogenation-deoxygenation section to ensure thorough deoxygenation and catalyst protection; while the hydrogen-to-oil ratio is moderately reduced in the isomerization section to optimize isomerization efficiency[119].

4.3. Effect of liquid hourly space velocity and reaction time

Both liquid hourly space velocity (typically expressed as liquid hourly space velocity, LHSV, defined as the volume of liquid feedstock processed per unit time per unit volume of catalyst) and reaction time have significant effects on the HEFA and CHJ, two processes for the production of Bio-jet fuel, but there are differences in the mechanism of the effects and the direction of optimization.

For the HEFA processes, the reaction time directly affects the depth of hydrocracking and product distribution. For example, at 350°C, the yield of Bio-jet fuel increases significantly from about 15% to 38% as the reaction time increases from 10 to 30 min[120]. The LHSV is a very critical parameter in the HEFA process, which determines the residence time of the feedstock in the catalyst bed and directly affects the degree of completion of the reactions such as hydrogenation, deoxygenation, isomerization and cracking[121]. Excessively high liquid hourly space velocity can result in insufficient contact time and incomplete conversion. A lower liquid hourly space velocity (i.e., longer residence time) typically facilitates adequate contact between reactants and the catalyst, promoting the reaction and enhancing the selectivity of the target product. However, excessively long residence times may cause overcracking, ultimately producing excessive light hydrocarbons (such as naphtha), reducing the selectivity of the jet fuel fraction (C_8-C_{16}), and adversely affecting catalyst lifespan. In industrial production, different liquid hourly space velocity and reaction times are typically set in stages. During the heavy oil desulfurization stage, the LHSV ranges from 1.0 to 1.5 h^{-1} (reaction time 10–60 min). For example, at a zero-rate LHSV of 1.0 h^{-1} on the Pt/Al-MCM-41 catalyst, the coal vapor ratio (aerosol/light hydrocarbons) reaches its peak. During the isomerization stage, LHSV ranges from 1.0 to 2.5 h^{-1} (reaction time 24–30 min). For instance, at 2.5 h^{-1}, the Pt/modified USY catalyst achieves jet fuel pour points below −50°C[84,122].

For the CHJ process, the reaction time in the first stage is an independent critical variable that requires precise control to balance the thorough depolymerization of biomacromolecules with the secondary repolymerization/coking of intermediate products. Insufficient retention time prevents the complete conversion of high-molecular-weight organic compounds in the waste into bio-crude oil, thereby reducing bio-crude oil yield. As the reaction time increases, bio-crude oil production significantly improves. Conversely, an excessively long reaction time causes intermediate products to undergo repolymerization, forming coke, or undergo gasification, producing CO_2, H_2, and other gases [123,124]. Research indicates that most liquefaction reactions can be completed within a relatively short period of time (e.g., 15–60 min) [111]. For example, a study determined that three hours was the optimal reaction time when converting palm oil into jet fuel fractions at 400°C [125]. Additionally, the effects of reaction time and temperature can be comprehensively evaluated using the reaction intensity factor (KSF). The highest biofuel conversion rate occurs when the KSF value ranges between 9 and 10[126]. In the second-stage hydrogenation upgrading process, to ensure complete reaction of complex components, removal of heteroatoms, and attainment of the required degree of saturation, a space-velocity range of 0.1 h^{-1} to 0.8 h^{-1} is typically employed for upgrading bio-crude oil[127].

Liquid hourly space velocity (or reaction time) is a key engineering parameter controlling reaction depth and chain reaction selectivity. Selecting an appropriate range within the process ensures sufficient contact between reactant molecules and active sites of the catalyst without inducing secondary cracking[128]. For the hydrothermal liquefaction stage of the CHJ process, an optimal reaction time window exists to balance the complete liquefaction of macromolecules with the undesirable repolymerization of intermediates. The core challenge in industrial design lies in maximizing the reactor volumetric efficiency by optimizing catalyst activity to achieve the highest possible liquid hourly space velocity, while maintaining the selectivity of the target products.

5. Conclusions and Outlook

5.1. Conclude

The conversion of WCO into SAF represents a technologically viable and strategically significant pathway for decarbonizing the aviation sector. This review has systematically examined the two most prominent technological routes, HEFA and CHJ, underscoring that catalyst design and process optimization are the central pillars determining efficiency, product quality, and economic feasibility. The mature and robust HEFA process relies on an advanced bifunctional catalyst system, while the emerging CHJ process offers greater feedstock flexibility and lower hydrogen consumption. However, it imposes stricter requirements on catalyst hydrothermal stability and process engineering.

A critical synthesis of the literature reveals that the evolution of catalyst systems, from conventional sulfide catalysts to advanced bimetallic, bifunctional, and molecular sieve-based designs—has been primarily driven by the need to enhance activity, selectivity, and robustness against the complex impurities inherent in WCO. The performance of these catalysts is inextricably linked to precise process conditions, including temperature, pressure, hydrogen-to-oil ratio, and liquid hourly space velocity, which collectively dictate reaction pathways, product distribution, and catalyst longevity. Despite significant progress, the transition from laboratory-scale excellence to robust, cost-

effective industrial operation remains hampered by challenges related to catalyst deactivation, the economics of hydrogen supply, and the integration of efficient pretreatment and product upgrading steps.

5.2. Challenges and prospects for the future

5.2.1. Technical challenges and responses

The pathway to large-scale commercialization of WCO-SAF faces persistent technical hurdles that demand focused innovation. Catalyst durability remains a primary concern. The complex impurity profile of WCO, including sulfur, chlorine, phosphorus, and trace metals, leads to irreversible poisoning and fouling of active sites. The current physical–chemical deacidification process has high energy consumption, while novel enzymatic deacidification (e.g., lipase-catalyzed ethanolysis) can reduce FFA to 0.25% under mild conditions at 40°C, but the large-scale application is still limited by enzyme cost and stability[43]. Furthermore, the high-temperature, high-pressure, and aqueous environments involved, particularly in CHJ, accelerate sintering, leaching, and support collapse. Future catalyst development must prioritize the design of "smart" materials with built-in resistance. This involves engineering robust supports (e.g., carbon-based materials, doped composite oxides), tailoring active phase compositions (e.g., forming stable alloys or intermetallic compounds), and constructing protective architectures like core–shell structures to physically and chemically shield active sites.

Process integration and energy efficiency present another major challenge. The HEFA process is energy- and hydrogen-intensive, with its carbon footprint heavily dependent on the source of hydrogen. The CHJ process, while less hydrogen-intensive, requires significant energy for maintaining high-pressure conditions. Responses should focus on developing low-severity catalytic processes, optimizing heat integration networks, and coupling SAF production with renewable hydrogen sources (green hydrogen) or in-situ hydrogen generation technologies to achieve true lifecycle carbon neutrality. Moreover, the development of robust, cost-effective pretreatment technologies to standardize highly variable WCO feedstocks is crucial for ensuring consistent catalyst performance and final fuel quality.

5.2.2. Prospects

Despite many challenges, technological innovation and policy drivers are accelerating the commercialization of SAF from WCO. In terms of catalyst stability enhancement, future research should be devoted to the development of multi-functional catalysts with low cost, high stability and anti-poisoning, especially non-precious metal catalysts and molecular sieve materials with graded pores. In terms of process, processes such as one-step hydrogenation and catalytic hydrothermal cracking need to be further optimized to reduce energy and hydrogen consumption, and focus on green-hydrogen coupling and reaction integration to enhance sustainability. Standardized pretreatment of feedstock is critical to ensure feed quality and process stability. Promoting the construction of pilot and demonstration plants with comprehensive techno-economic and full life-cycle assessments is key to validating their commercialization feasibility and gaining policy support.

Furthermore, the integration of WCO-SAF production with other waste valorization streams (e.g., biogas upgrading, municipal solid waste processing) within a biorefinery concept can enhance overall economics and sustainability. From a policy and market perspective, the strengthening of mandates like ReFuelEU and the development of robust carbon accounting and trading mechanisms will provide stable demand signals, accelerating investment and scale-up. As technological readiness advances and costs decline, WCO-SAF is poised to transition from a niche supplement to a mainstream component of the global aviation fuel mix, playing an indispensable role in the sector's journey toward net-zero emissions.

CRediT authorship contribution statement

Zifan Huang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Data curation. **Lang Chen:** Visualization, Investigation, Data curation. **Liuxia Li:** Methodology, Investigation. **Jinglin Li:** Investigation, Formal analysis. **Pan Wang:** Writing – review & editing, Validation, Supervision, Project administration, Conceptualization. **Lianhai Ren:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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